

GAME ANALYTICS: UNLOCKING TENNIS INSIGHTS

An End-to-End Data Engineering &
Predictive Analytics Suite



OUR TEAM



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PROJECT VISION & OBJECTIVES

The Vision: To transform complex, unstructured live sports data into a centralized decision-making engine for players, organizers, and sponsors.

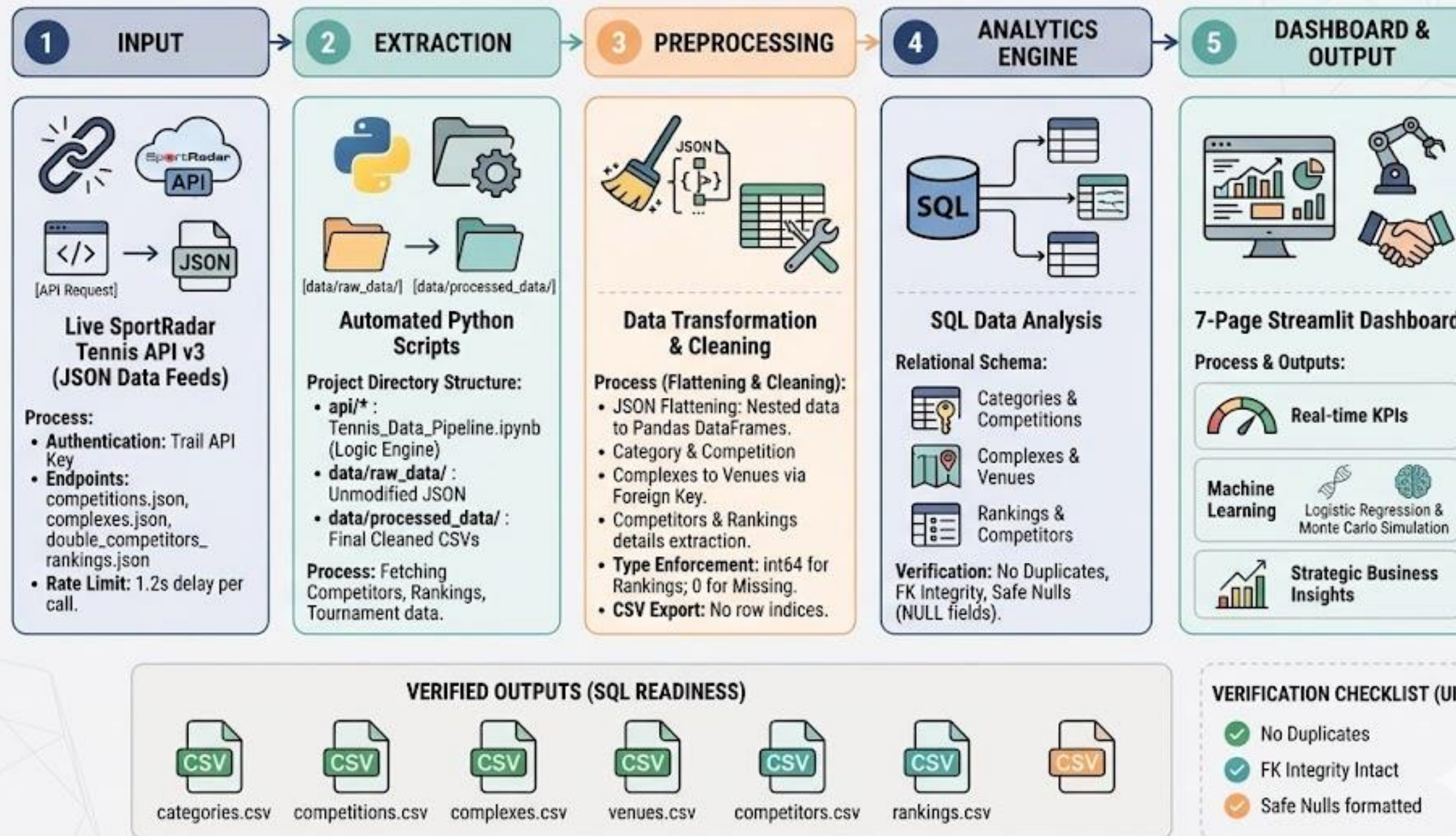
- **Key Objectives:**

- 1. Automate:** Streamline live data ingestion from SportRadar APIs.
- 2. Analyze:** Implement a robust SQL database for multi-dimensional querying.
- 3. Predict:** Use Machine Learning to simulate match outcomes with deterministic accuracy.
- 4. Visualize:** Deploy an interactive suite of analytics tools for non-technical stakeholders.

GRAPHICAL ABSTRACT

SYSTEM WORKFLOW

Tennis Game Analytics System Workflow



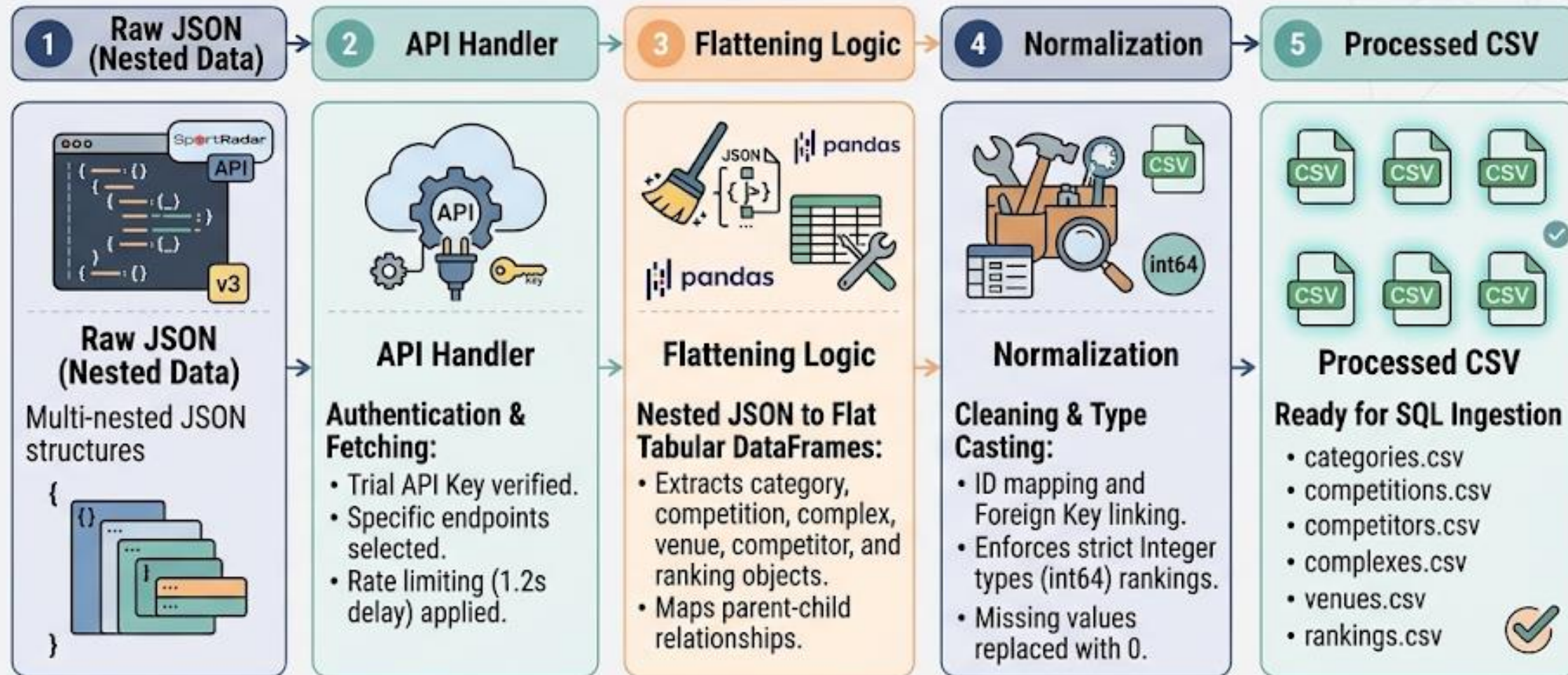
TECHNICAL STACK

- **Data Layer:** SportRadar API (Live Data), JSON, CSV.
- **Data Engineering Layer:** Python libraries (Pandas, NumPy, Requests).
- **Storage Layer:** SQL Database (Relational Schema: Competitors, Venues, Rankings).
- **Intelligence Layer:** Scikit-Learn (ML Models), Monte Carlo Simulation.
- **User Layer:** Streamlit (Framework), Plotly (Interactive Charts).

DATA ENGINEERING PIPELINE

TENNIS GAME DATA ENGINEERING PIPELINE

Tennis Game Data Engineering Pipeline



PIPELINE CONTEXT & DATA FLOW



SQL DATA ANALYSIS: CATEGORIES & COMPETITIONS

Query 1: competitions volume per category

```
SELECT c.competition_name, cat.category_name  
FROM competitions c  
JOIN categories cat ON c.category_id = cat.category_id;
```

- **Output:** Provides a clear mapping of competitions to their respective categories such as ATP, WTA, and ITF.
- **Insight:** **ATP** dominates the event hierarchy, indicating higher market value.

RECOMMENDATION: FOCUS SPONSORSHIP AND MARKETING RESOURCES ON **ATP SINGLES** EVENTS FOR MAXIMUM ROI.

Query 2: Distribution of Competition Types

```
SELECT type, COUNT(*) FROM competitions  
GROUP BY type;
```

- **Output:** Singles (350), Doubles (120), etc.

Insight: Singles matches are the primary revenue drivers.

SQL DATA ANALYSIS: COMPLEXES & VENUES

Query 1: Venue Density per Complex

```
SELECT complex_name, COUNT(*) as total_venues  
FROM venues  
GROUP BY complex_name  
ORDER BY total_venues DESC;
```

- **Output:** Provides a clear mapping between venues and their parent complexes.
- **Recommendation:** Use this relationship to organize venue-level data within larger complexes in the dashboard.

Query 2: Global Infrastructure Reach

```
SELECT country_name, COUNT(*) FROM venues  
GROUP BY country_name;
```

- **Output:** Shows which countries have strong infrastructure presence.
- **Recommendation:** Focus expansion and investment in regions with fewer venues.

SQL DATA ANALYSIS: RANKINGS & COMPETITORS

Query 1: Top 5 Elite Competitors

```
SELECT c.name, r.ranks
FROM rankings r
JOIN competitors c ON r.competitor_id =
c.competitor_id
WHERE r.ranks <= 5
ORDER BY r.ranks;
```

Output: Top-ranked players attract the most audience attention and engagement

Recommendation: Highlight these players in dashboards, promotions, and marketing campaigns.

Query 2: Number of competitors per Country

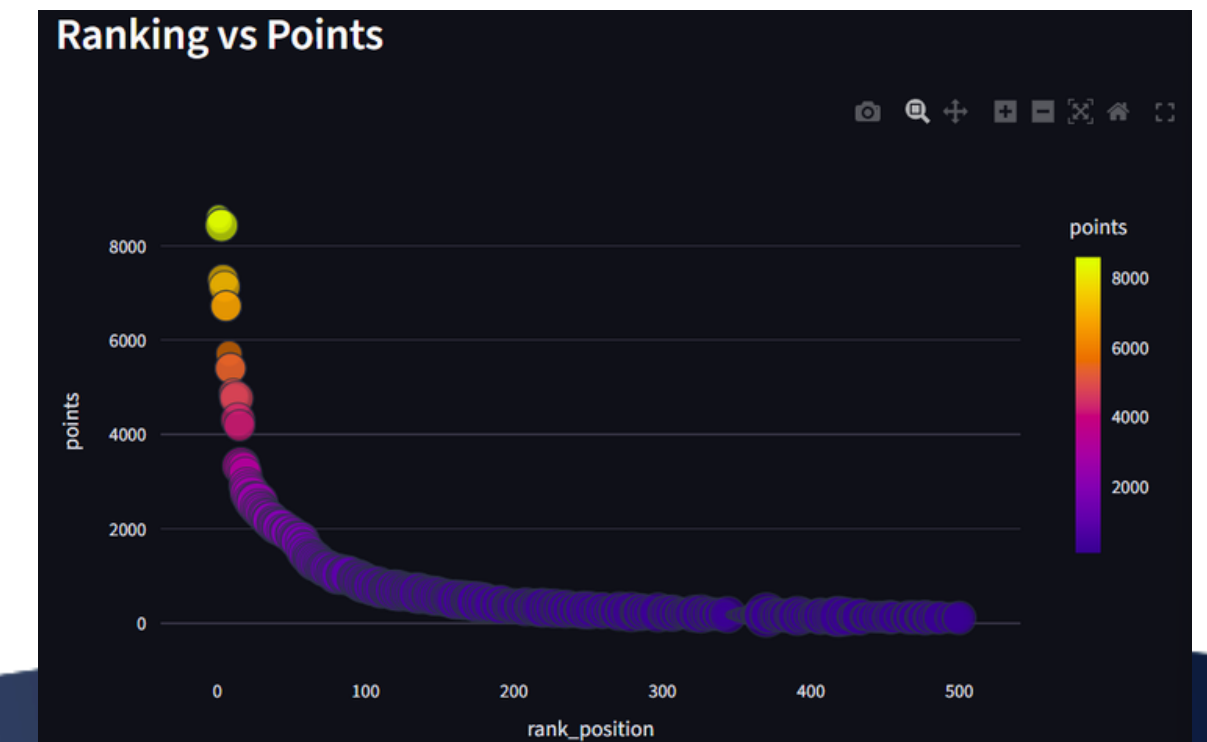
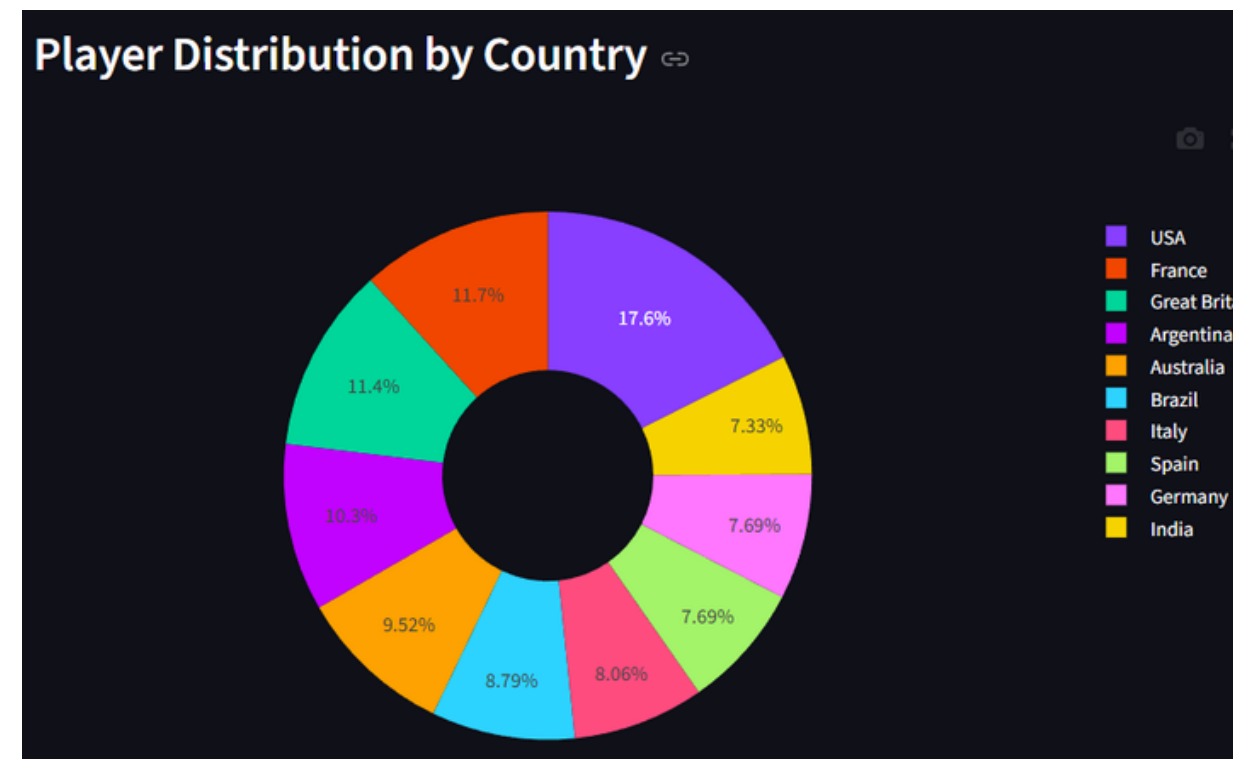
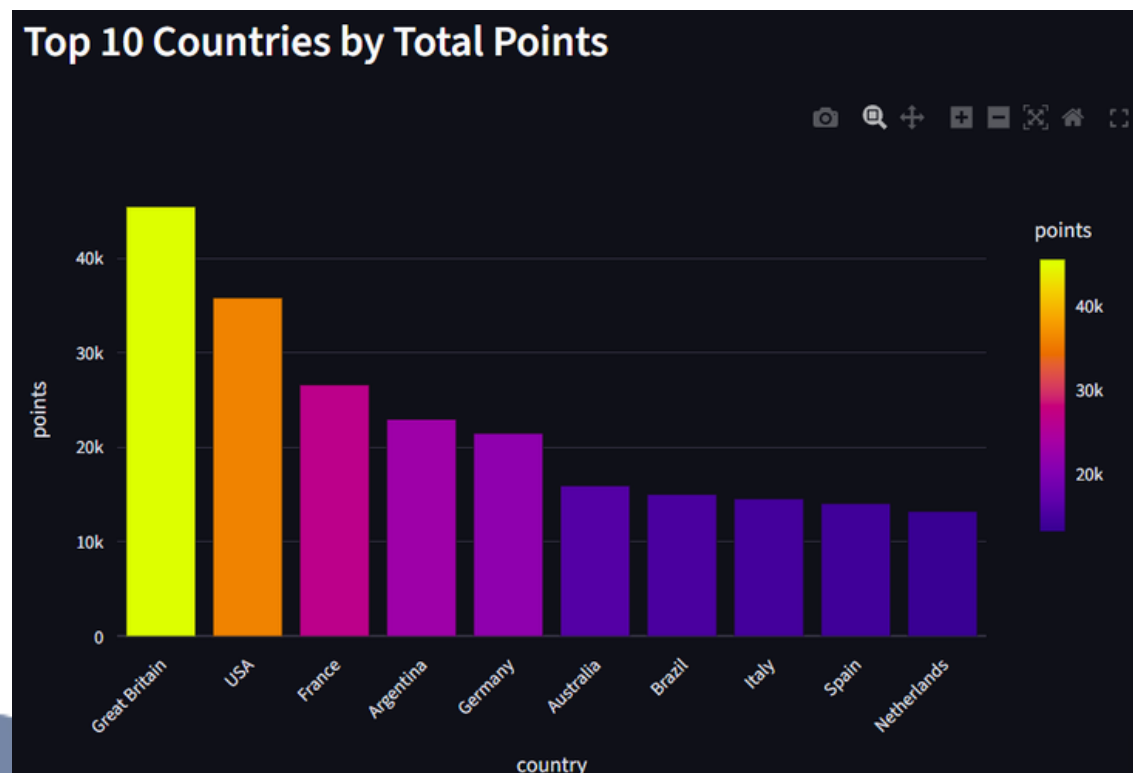
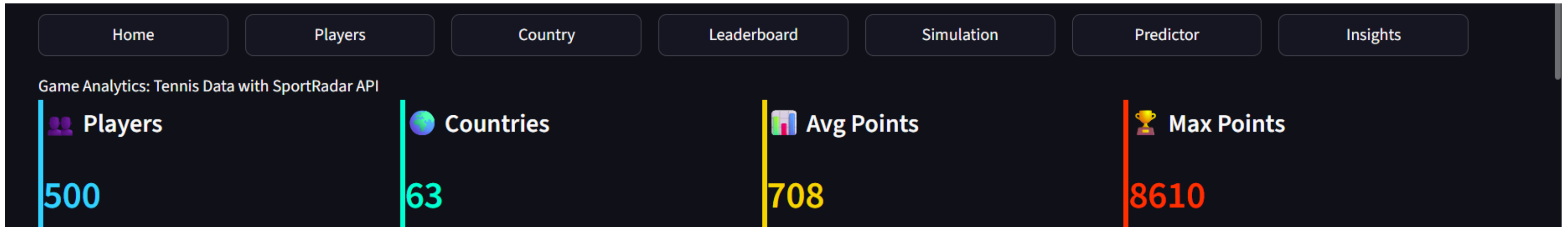
```
SELECT country, COUNT(*) AS total_competitors
FROM competitors
GROUP BY country
ORDER BY total_competitors DESC;
```

Output: Shows the distribution of talent across different countries

Recommendation: Encourage participation and talent development in countries with lower representation.

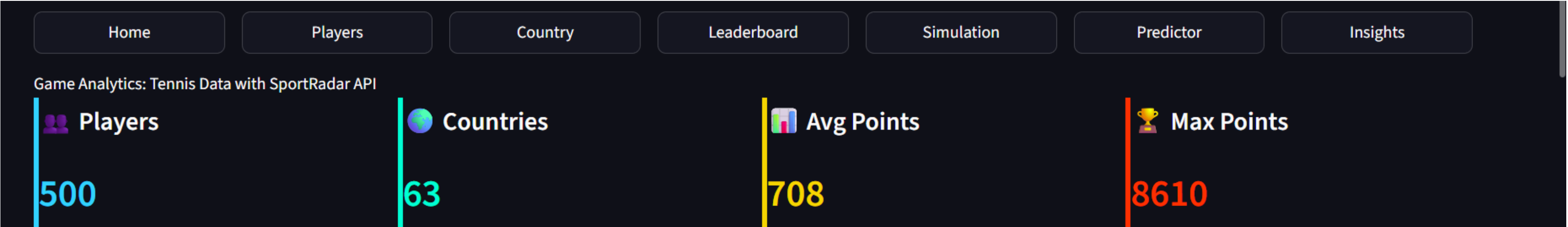
DASHBOARD PAGE HIGHLIGHTS

1.Home: The Home Dashboard is designed to provide a comprehensive overview of the entire dataset. It acts as the entry point of the application and enables users to quickly understand the scale, distribution, and performance trends of tennis players.



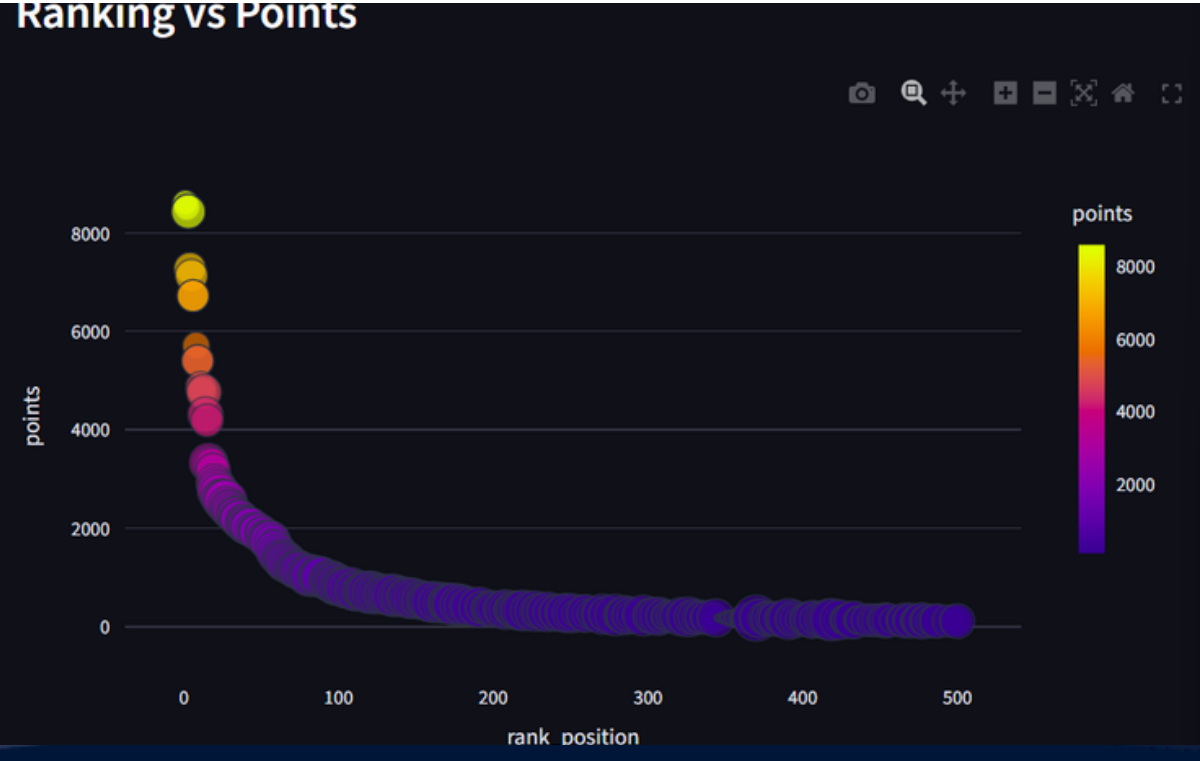
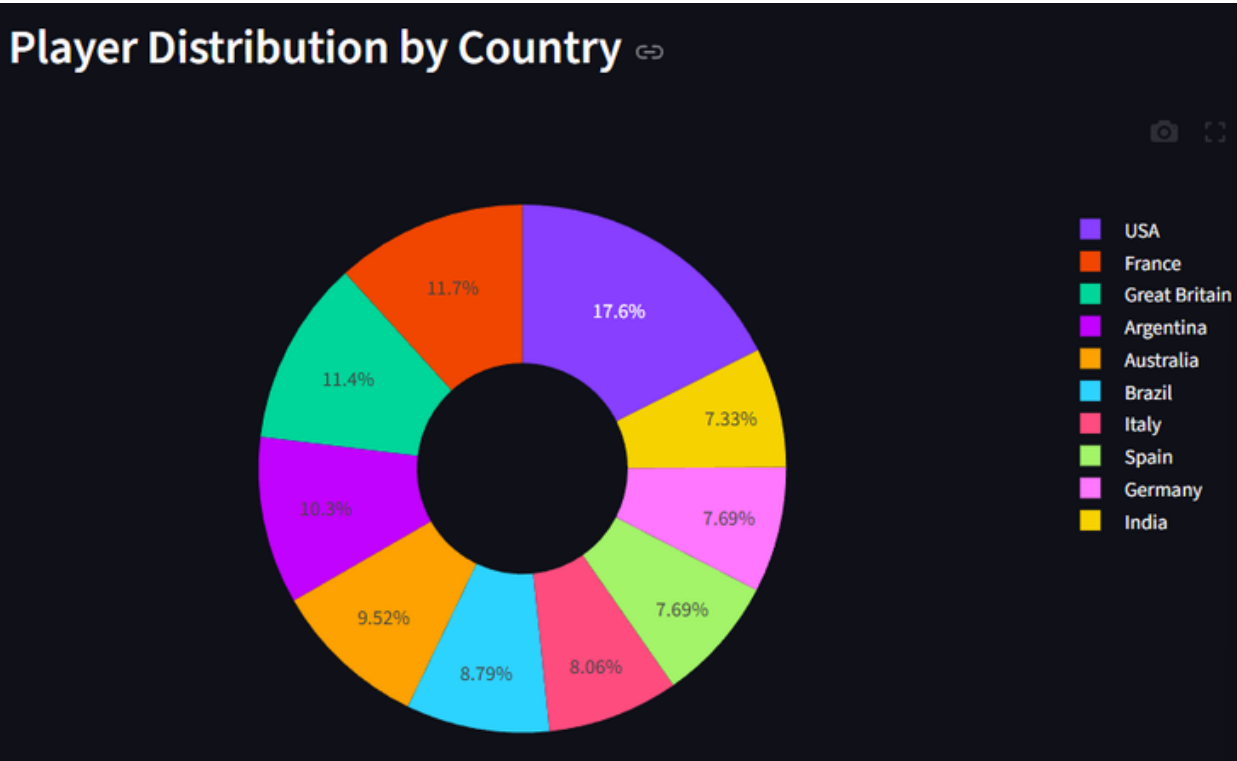
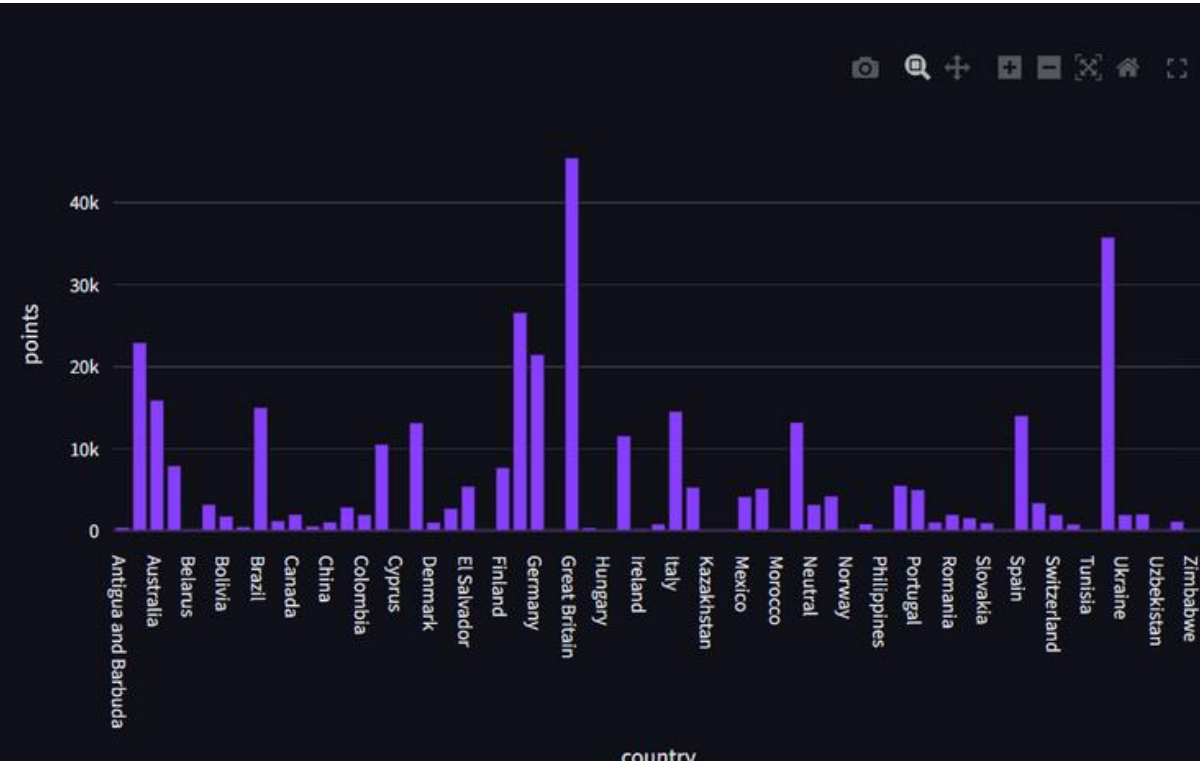
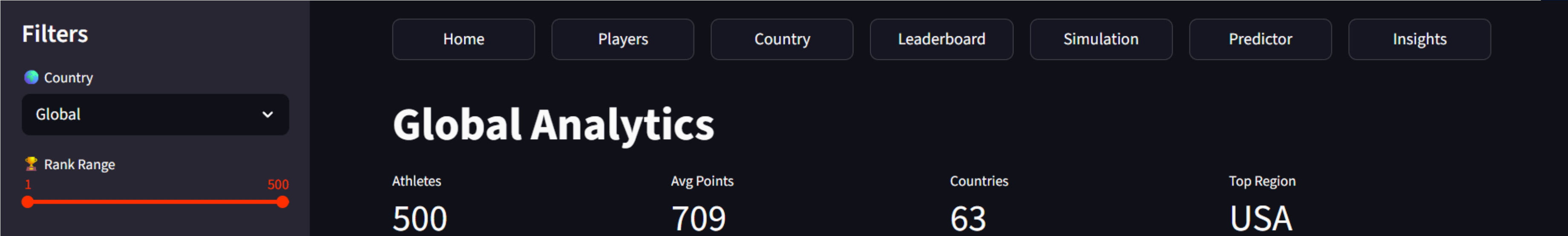
DASHBOARD PAGE HIGHLIGHTS

2.Players Deep Dive Page : This page focuses on a detailed analysis of individual players. It allows users to explore player-specific metrics and evaluate performance.



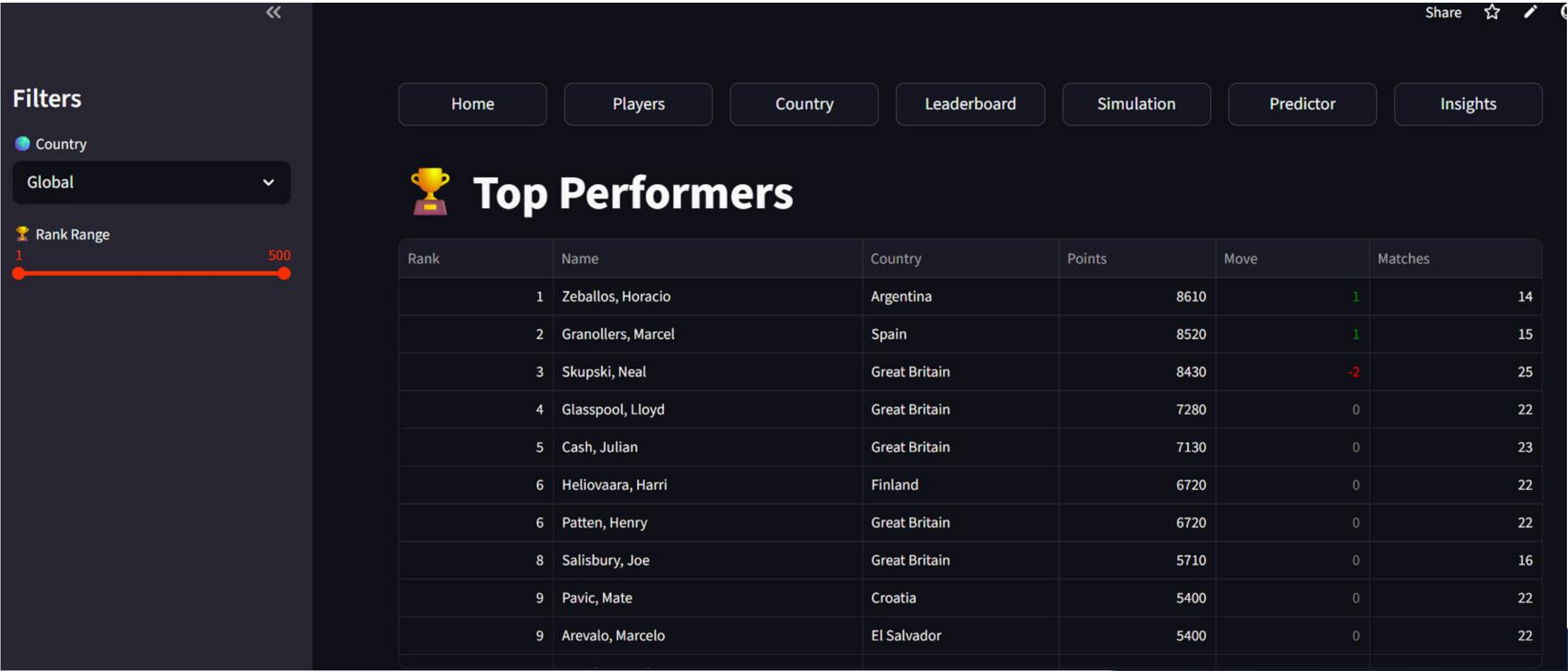
DASHBOARD PAGE HIGHLIGHTS

3.Global Analytics Page: To provide a macro-level view of tennis performance across countries.



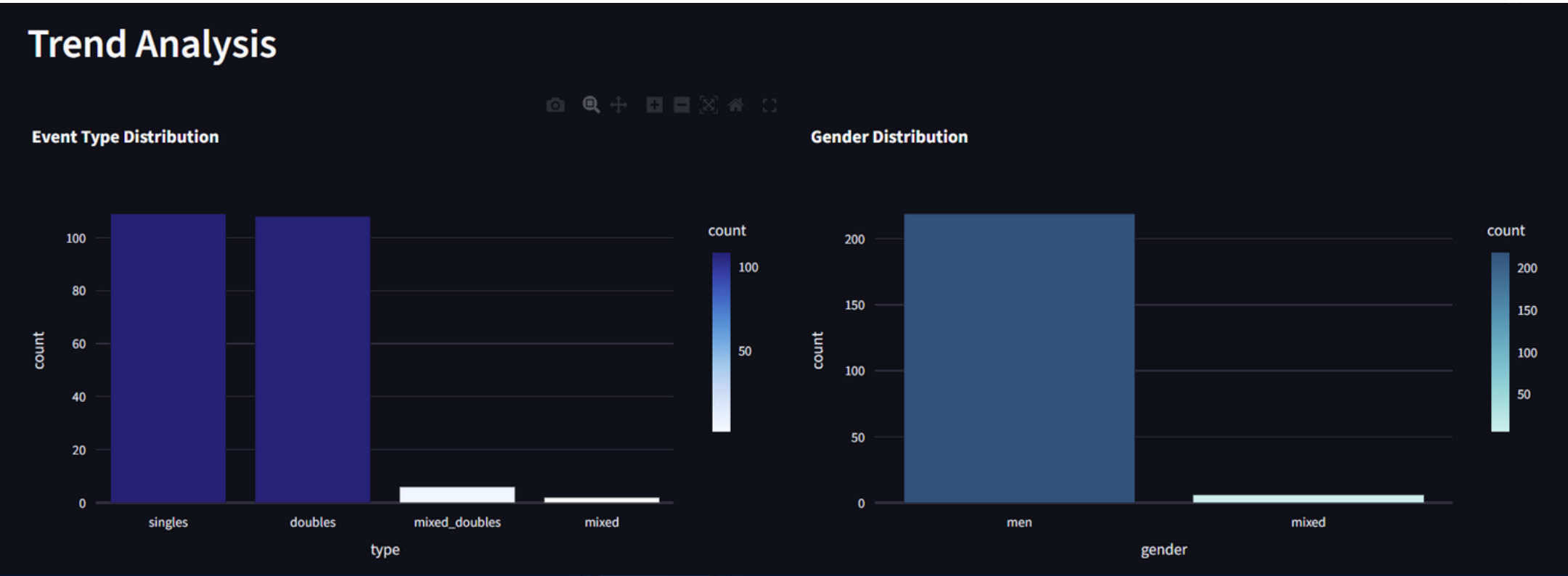
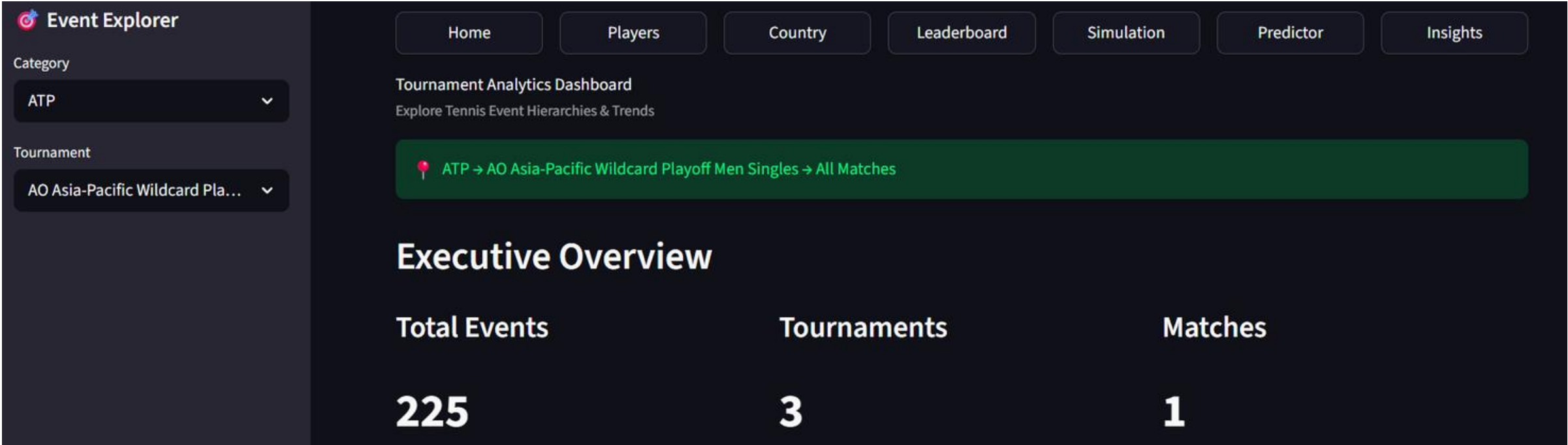
DASHBOARD PAGE HIGHLIGHTS

4. Leaderboard Page : To display ranked players in a structured table format.



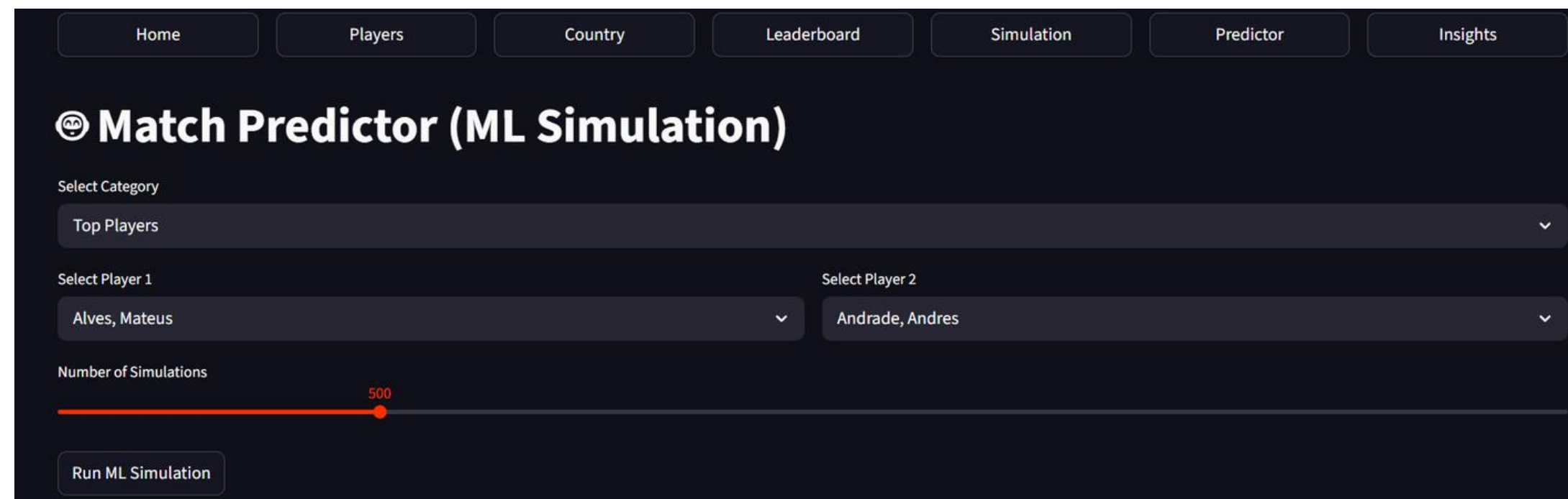
DASHBOARD PAGE HIGHLIGHTS

5. Event Explorer Page : Analyzing tournament hierarchy and gender/event-type distribution (Singles vs. Doubles).

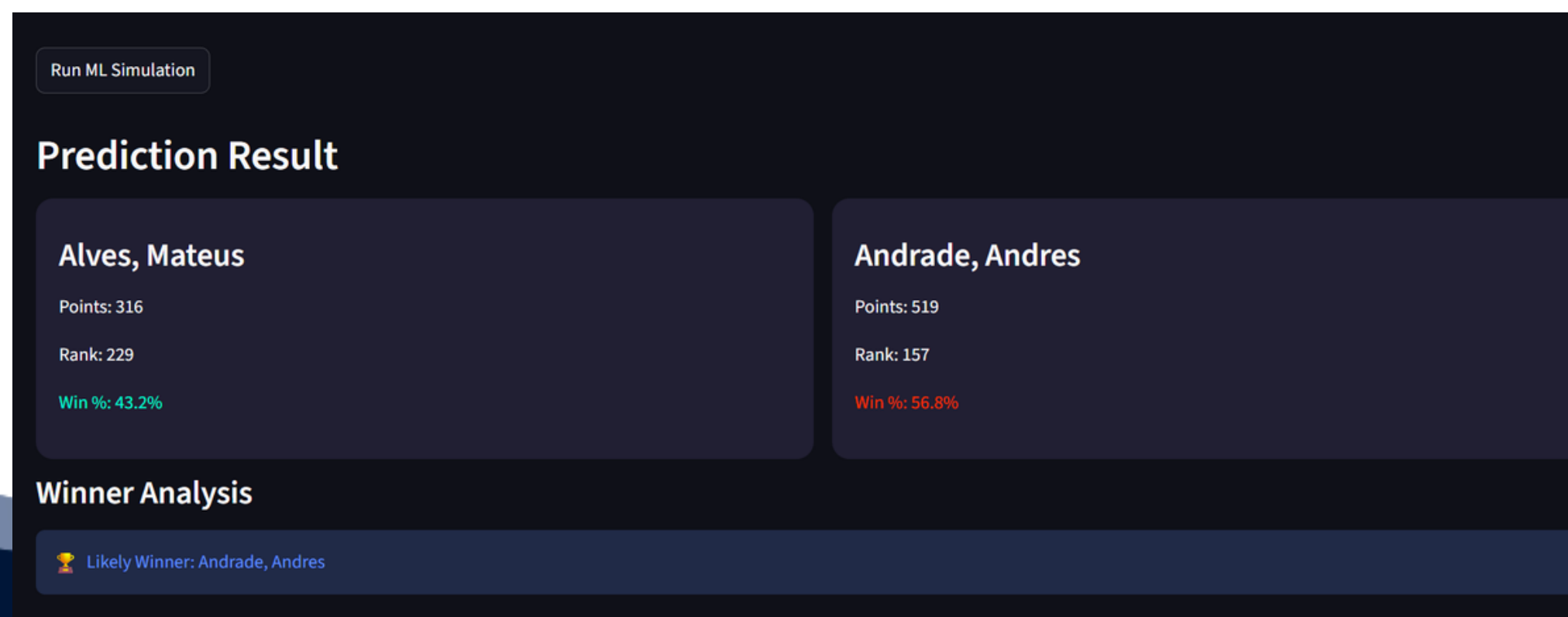


DASHBOARD PAGE HIGHLIGHTS

6. Match Predictor : Advanced "Match Simulation" tool using ML to convert player data into actionable match probabilities.

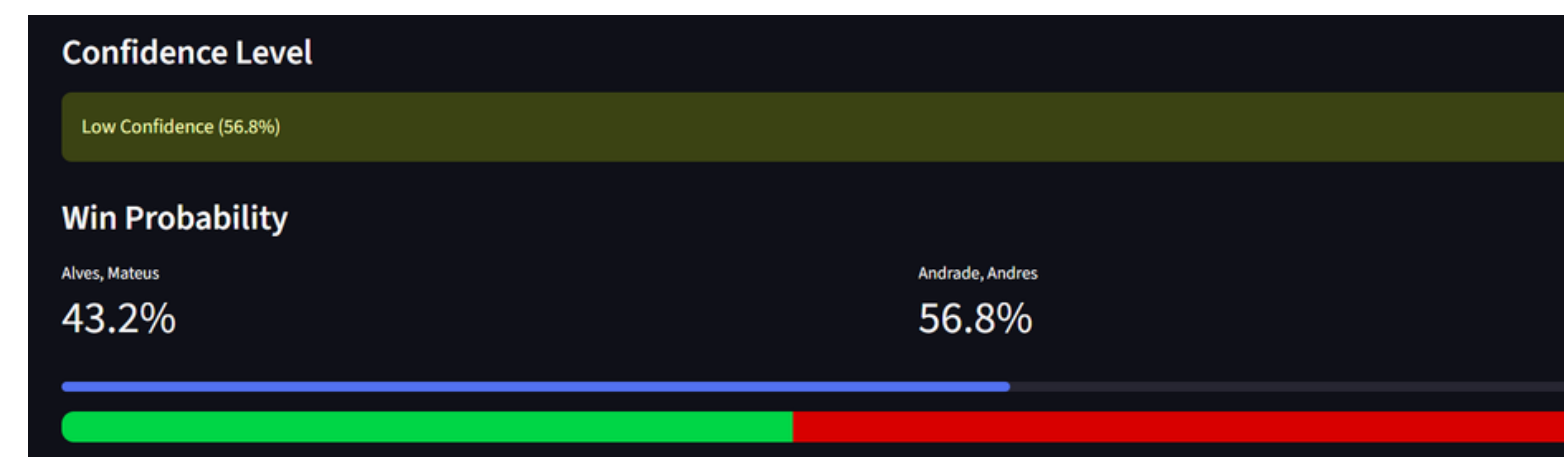


The image shows a web form titled "Match Predictor (ML Simulation)". At the top, there is a navigation bar with buttons for Home, Players, Country, Leaderboard, Simulation, Predictor, and Insights. The form itself has a dark theme. It starts with a "Select Category" dropdown menu set to "Top Players". Below this are two "Select Player" dropdown menus. "Select Player 1" is set to "Alves, Mateus" and "Select Player 2" is set to "Andrade, Andres". Underneath these is a "Number of Simulations" slider, which is currently set to 500. At the bottom of the form is a "Run ML Simulation" button.



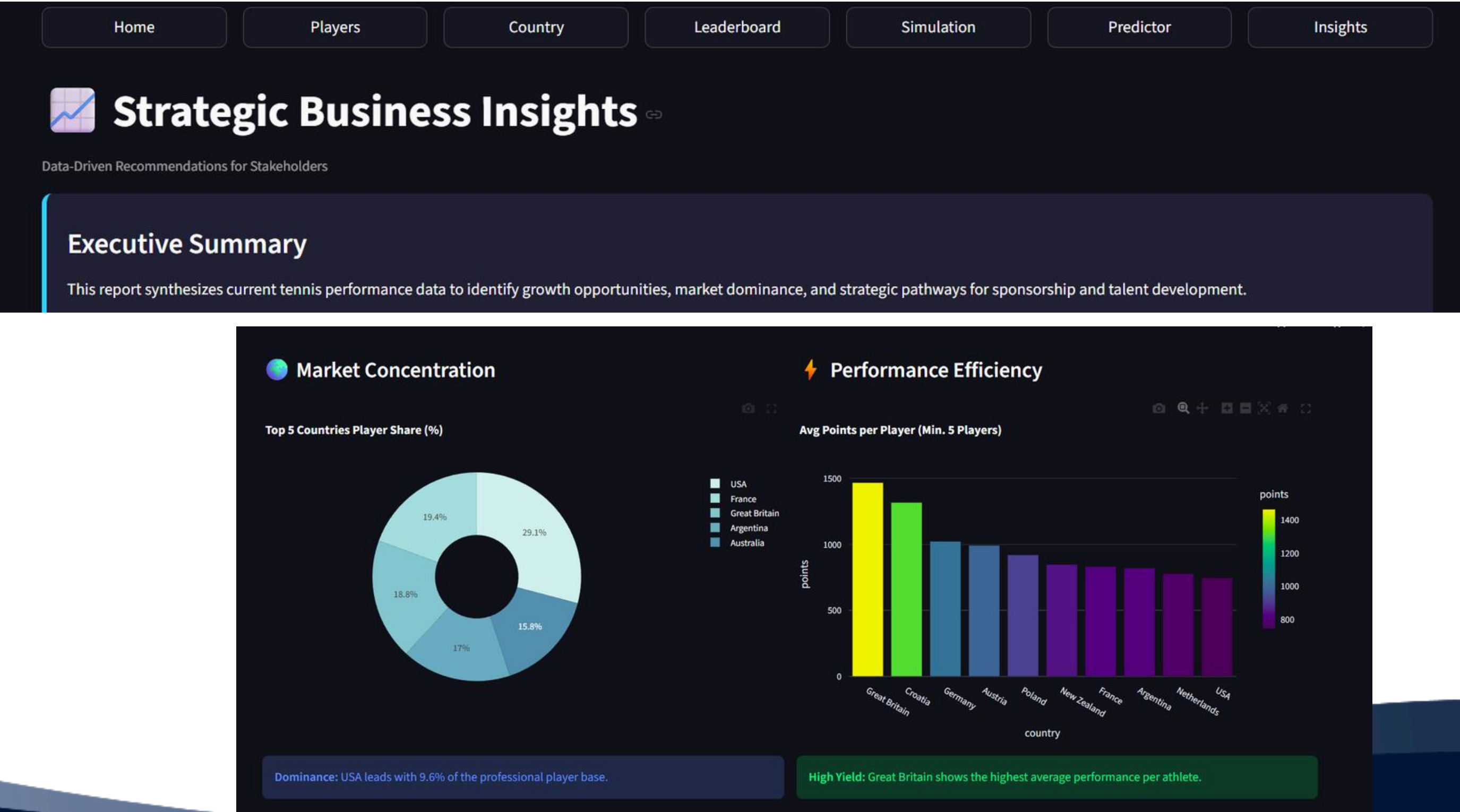
This section displays the results of the ML simulation. It features a "Run ML Simulation" button at the top left. The main heading is "Prediction Result". Below this, there are two player profiles side-by-side. The left profile is for "Alves, Mateus" with 316 points, rank 229, and a 43.2% win probability. The right profile is for "Andrade, Andres" with 519 points, rank 157, and a 56.8% win probability. Below the player profiles is a "Winner Analysis" section which states "Likely Winner: Andrade, Andres".

Player	Points	Rank	Win %
Alves, Mateus	316	229	43.2%
Andrade, Andres	519	157	56.8%



DASHBOARD PAGE HIGHLIGHTS

- **7. Strategic Business Insights Page** : Translates technical data into business strategies.



DASHBOARD PAGE HIGHLIGHTS

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Growth Opportunities



Emerging Markets

Countries with high performance growth but low current player volume.

- **Ghana:** +30.0 avg rank jump
- **China:** +13.5 avg rank jump
- **Ukraine:** +12.3 avg rank jump



Event Portfolio

The circuit is currently **53.6% Singles** dominated.

Recommendation: Increase investment in Doubles and Mixed formats to diversify audience engagement.



Sponsorship Targets

Top 'Fastest Risers' this week:

- **Dostanic, Stefan** (USA)
- **Rybakov, Alex** (USA)
- **Casanova, Hernan** (Argentina)



Strategic Recommendations

1. **Regional Diversification:** While Europe remains the stronghold, data shows high 'movement' trends in emerging regions. Allocating scouting resources to these areas could yield high ROI.
2. **Performance-Based Tiering:** Implement tiered sponsorship models based on the 'Points Capacity' metric visualized in the Player Deep Dive.
3. **Tournament Scheduling:** Align event promotion with the geographic distribution of 'Top 100' players to maximize local viewership and ticket sales.
4. **Digital Engagement:** Leverage the 'Match Predictor' logic for fan engagement platforms, as predictive analytics are a key driver for younger demographics.

FUTURE SCOPE

- **Real-time Integration:** Direct live-streaming data feeds for in-play predictions.
- **Advanced Models:** implementing XGBoost or Deep Learning (LSTM) for point-by-point live odds.
- **Health & Fatigue Tracking:** Integrating player injury data and "Time on Court" metrics for more accurate simulations.
- **Cross-Sport Portability:** Scaling the pipeline architecture for Cricket, Football, and F1.



CONCLUSION

In summary, this project isn't just about tennis rankings, it's about building a robust engine that turns massive volumes of sports data into a competitive advantage. We have created a tool that doesn't just look at what happened in the past, but predicts what will happen in the future.

**QUESTIONS? WE'D
LOVE TO hear
them.**

THANK
you